

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output content (label correctness for the deployment context) is the dimension most exposed to validity violations. The 96% spot-check agreement [Q43, Q44] was conducted on 100 randomly sampled questions [Q42] without country stratification, so it cannot speak to per-country annotation quality. Annotators are exclusively from Jordan, Egypt, Lebanon, UAE, and KSA [Q31] — Morocco, Kuwait, and Palestine have no documented annotators [Q96], yet Palestine is a top-three question contributor [WEB-1] and Moroccan law content is present [DATASET-D34]. This means Palestinian, Moroccan, and Kuwaiti questions (where present) had ground-truth labels verified by workers from other national educational contexts. Empirically, the dataset analysis shows: (a) Jordanian-curriculum facts (King Talal's reign, 1952 constitution, Royal Medical Services, CBJ URL) are encoded as universally correct answers [DATASET-D3, D4, D7, D20, D21]; (b) Egypt-specific currency and institutions are encoded as universal in university economics [DATASET-D28, D39, D40]; (c) Moroccan procedural law is encoded as 'the' law answer without jurisdiction tag [DATASET-D8, D9, D10, D34]; (d) at least one likely-incorrect ground-truth label is observed [DATASET-D32]; (e) OCR-corrupted questions [DATASET-D14, D15] cannot have reliably verifiable answer keys. Pretraining contamination is also acknowledged as not ruled out [Q94, Q95]. For a deployment that requires curriculum-aligned answers per country [elicitation A1] and country-specific legal correctness [elicitation A2], the ground-truth labels do not reliably reflect regional stakeholder perspectives for Moroccan, Emirati, Kuwaiti, Saudi, Lebanese, or Palestinian cohorts.

ID	Evidence
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q42	"Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q44	"This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)
Q94	"It is important to emphasize that our experimental results do not provide conclusive answers regarding the performance of LLMs in Arabic." (p.9)
Q95	"to a lack of sufficient information about its training regimen. As such, we cannot assert that the model's pretraining data is free from contamination." (p.10)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
D3	"ب وه ةفأك ةلودل نوؤش ىلع ةماعل ةيالول بحاص سلجلم" with answer "عرازول سلجم" — "The council with general sovereignty over all state affairs is the Council of Ministers" — specifically Jordanian constitutional structure
D4	"Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom
D7	0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact
D8	— "ىوعدلا رهوج يف مكحل رودص دعب الإ عوفد وأ ضراع عازن يف ةرداصل وأ ةيديهمتمل ماكحلأ فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure

D9	عافدل اب ةفل كملم ةيموكحل اطلسل او لدعل ربزو نم رارب...نيكي كدلل ةيئاضقلا ططرشل ا طباض ةقص لي وخت نكمي Moroccan judicial police powers — references "يكلملا كدلا" (Royal Gendarmerie of Morocco) — ينطولا
D10	Moroccan civil party withdrawal in criminal proceedings — عيحص ةبوجل ا عيجم: مكحل رودص لبقي ندم فرطلا لزانن اذ
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source — لري تدت...ةتفل تخسل ا تيرادل ا تايهت تدسل او صاختل ال ني تب ةطلتل ا عتييهت ةتجرد
D15	Severely OCR-corrupted management question with unreadable answer options — ت ح... ت محرخل... امدختسل ا ةسئاد ططخل ا عل ل اثم دعي ل
D20	"Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services — ةي بطلا ةني دمل ا يفشتسم: انه... تاسسؤملا نم ديدعل ا ةيموكحل ا ةيكلمل ا ةي بطلا تامدخلا لمشت
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school — ةنس همكح ةدمو، 1952: يف كل ذو ةيحص بابسل ا هتالو تهتن ني سحل نب لال طلكلم
D28	Egyptian currency (جني = 100 Egyptian pound) in university economics — Egyt-specific monetary context — وه ةرمثتسملا لاول امل ا عل ةئاعلا لدعم...هني ج 100 بولطم رامثتسال ا مجح ناك اذ
D32	تايمكب هرفاوت: ل ا كل ذ يف ببسل ا عجي...ي عي بطلا زاغلا و طفنلا نم ةريكب تاوثرثب ةيبرعل ا نبال ا عل هلل ا عان "Gas is available in limited quantities" — possibly incorrect answer key for general Arab world social science — قليل
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)

Strengths

- Annotator credentials are documented: Master's students/Research Assistants and Bachelor's-degree-holders, all native MSA speakers [Q35, Q39]
- 96% inter-annotator agreement on 100-question spot check provides a baseline quality signal [Q43]
- Compensation exceeded monthly average wages in respective countries [Q36]
- One-hour orientation workshop conducted before data collection [Q40]
- STEM ground truth (math, biology, physics, factual CS) is curriculum-neutral and reliable across all eight countries [DATASET-D24, strength 5]

ID	Evidence
Q35	"Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)
Q36	"We ensured competitive compensation for the workers, exceeding the monthly average wage in each respective country." (p.4)
Q39	"While our workers are native speakers of Modern Standard Arabic with at least Bachelor's degrees, we maintain the quality of our dataset construction through meticulous steps." (p.4)
Q40	"Firstly, we conducted a 1-hour workshop before the data collection stage to clarify the process." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
D24	اذا جتن اذ — اهتيبثت مت يتي ال (CO2) تائي زج ددع مكف نفل اك ةقلح نع زوكولغ تائي زج (3) جتن اذ Palestine-sourced biology; universal factual content

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?

OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels reflect regional stakeholder perspectives unevenly. For STEM and curriculum-neutral content, labels are reliable across all eight countries. For civics, history, social science, economics, and law, labels reflect the originating exam's jurisdiction without country-stratified validation [DATASET-D1, D7, D34].

OC-2: Disagreement is highly likely between original annotators and regional populations for: (a) Moroccan students reviewing Jordanian civics [DATASET-D1, D2]; (b) UAE/Kuwait students reviewing Egyptian university economics [DATASET-D28]; (c) Egyptian/Saudi/Jordanian/UAE/Kuwaiti/Lebanese/Palestinian law students reviewing Moroccan procedural law [DATASET-D8, D9, D10]; (d) Palestinian students reviewing Jordanian-narrative King Abdullah I content [DATASET-D31].

OC-3: Annotator demographics documented [Q31] but limited: 10 annotators total, all from Jordan/Egypt/Lebanon/UAE/KSA. No Morocco/Kuwait/Palestine representation [Q96]. No formal datasheet or data statement beyond paper text.

OC-4: Re-annotation by representative annotators from Morocco, Kuwait, and Palestine is essential before deployment use, especially for civics, history, law, social science, and economics subsets.

OC-5: Aggregation: 96% agreement reported as a single global figure [Q43] without country stratification — minority-country perspectives are erased in the aggregate. No formal IAA statistic beyond this [Q43].

OC-6: Documented label issues: (a) Jordanian-curriculum facts as universal ground truth [DATASET-D1, D3, D4, D7, D20, D21]; (b) Egyptian university content as universal [DATASET-D28, D39, D40]; (c) Moroccan-only law ground truth [DATASET-D34]; (d) no Palestinian annotator review of Palestinian content [Q96, WEB-1]; (e) at least one likely-incorrect answer key observed [DATASET-D32]; (f) OCR-corrupted questions where verifiability is compromised [DATASET-D14, D15]; (g) potential pretraining contamination not ruled out [Q94, Q95].

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D1	What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics
D2	The first coalition government in Jordan in 1952 was known as the ___ government — Jordanian political history
D3	The council with general sovereignty over all state affairs is the Council of Ministers — specifically Jordanian constitutional structure
D4	Rights under the [Jordanian] Kingdom's constitution — explicitly references the Jordanian Hashemite Kingdom
D7	0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact

D8	— يوعدل رهوج يف مكحل رودص دعب ال عوفد وأ ضراع عازن يف ةرداصل وأ ةيدي مهمتلا ماكحلأ فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
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D20	"Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services — ةي بطل ةني دمل ىف شتسم :اهنم...تاسس ؤمل نم ديدعل ةيموكحل ةيكللمل ةي بطل تامدخل لم شت
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school — ةنس همكح ةدمو ، 1952م :في كل ذو ةي حص بابسأل هتيلو تهنتا نيسحل نب لال طلكلمل
D28	Egyptian currency (ج = % ١٥ :وه ةرمثتسم لم لاومأل ىل ع دئاعل لدعم...هي ن ج ١٠٠٠٠٠ بولطم رامثتسال م ج ج ناك اذا Egyptian pound) in university economics — Egypt-specific monetary context
D31	King Abdullah I's positions on Zionism — highly sensitive political content, Jordanian framing of Palestinian history — ني طسلف في اعامطأ ةي نوي هصل ةكرحل نم لوأل هللدبع لكلمل فق او نم تسيل ةي تالآ نم ةدحاو
D32	تاي مكب هرفاوت :ىل لكذ في ببسل عجري...ي عي بطلا زاغال و طفلن نم ةري بك تاو رثب ةي برعل اندال ب ىل عل هلل مع أن "Gas is available in limited quantities" — possibly incorrect answer key for general Arab world social science — ةلي ق
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Information Gaps

- Country-stratified inter-annotator agreement is not reported [Q43]
- Pretraining contamination extent is not measured [Q95]
- Total proportion of incorrect answer keys beyond the 4% noise floor implied by 96% agreement is unknown

Requires Expert Verification

- Per-country curriculum experts (especially Moroccan Maliki Islamic studies, Kuwaiti civics, Palestinian national history, UAE 4-4-4 curriculum) to re-validate ground-truth labels
- Legal experts from Egypt, UAE, KSA, Jordan, Lebanon, Kuwait, Palestine to assess law-subset jurisdictional correctness

Remediation

Gap 1: Ground-truth labels for Palestinian, Moroccan, and Kuwaiti questions were verified by workers without those national curriculum backgrounds [Q31, Q96]; 96% agreement is a global figure not stratified by country [Q43].

Recommendation: Recruit at least 2–3 curriculum experts each from Morocco, Kuwait, and Palestine to re-verify ground-truth labels for at least the civics, history, social science, and law subsets relevant to those countries. Report country-stratified IAA before deployment use.

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